

The Epistemology of AI Doom: A Rigorous Critique of the Existential Risk Chain

The modern discourse surrounding artificial intelligence is increasingly dominated by a singular, dramatic hypothesis: the proposition that unchecked advancements in machine learning will inevitably result in the extinction or irreversible subjugation of humanity. This existential risk (x-risk) argument has moved from the fringes of speculative philosophy into the core of mainstream technological governance, fundamentally altering the trajectory of international regulatory frameworks, corporate governance, and academic research priorities. The intellectual architecture of this argument largely originates from the theoretical work of Nick Bostrom, Eliezer Yudkowsky, and institutions such as the Machine Intelligence Research Institute (MIRI), the Center for AI Safety (CAIS), and the broader Effective Altruism (EA) ecosystem. More recently, empirical researchers such as Geoffrey Hinton have adopted variations of this concern, and popularizers like Robert Miles have widely disseminated its core tenets to the public. One of the most persistent analytical failures in public and academic discussions of AI x-risk is the tendency to treat the hypothesis as a monolithic, indivisible claim. In reality, the x-risk argument is a sequential causal chain, characterized by a series of distinct, contingent propositions. The strongest and most formalized version of the argument proceeds as follows:

1. Current machine learning (ML) architectures will scale into Artificial General Intelligence (AGI).
2. AGI will scale into superintelligence via recursive self-improvement.
3. Superintelligence will inherently tend toward power-seeking behavior.
4. Power-seeking behavior will result in a catastrophic loss of human control.
5. This loss of control will culminate in existential catastrophe.

In the domain of formal logic and probability theory, the probability of a conjunctive chain of independent events is the product of their individual probabilities. Therefore, the certainty of the final conclusion relies on the robust empirical and theoretical validation of every preceding link. The following analysis rigorously decouples these propositions, subjecting each step to exhaustive scrutiny through the lenses of cognitive science, endogenous growth economics, philosophy of mind, international relations, and decision theory.

Critique 1: The Inductive Leap from Current ML to AGI

The foundational premise of the x-risk chain posits that current dominant paradigms in generative artificial intelligence—specifically the massive scaling of generalist large language models (LLMs)—will eventually yield Artificial General Intelligence. This proposition is fundamentally an induction problem that leaps from impressive capabilities to general cognitive equivalence.

The empirical support for the transition from ML to AGI rests heavily on scaling laws, which

document predictable, proportional gains in model performance as functions of increased parameter count, training data, and computational power. LLMs have demonstrated remarkable emergent capabilities, including transfer learning, tool use, reasoning-like behavior, and complex code generation¹. However, the inference that these capabilities equate to a trajectory toward general intelligence is logically perilous. Historically, scaling a technology does not alter its fundamental paradigm: airplanes do not become birds by scaling, calculators do not become mathematicians by scaling, and chess engines did not spontaneously develop general intelligence through increased compute¹.

Critics of the scaling hypothesis, notably Yann LeCun and Gary Marcus, argue that LLMs represent a local optimum rather than a continuous path to AGI¹. Current architectures exhibit profound limitations in domains requiring persistent world models, grounded understanding, autonomous planning, and robust causal reasoning⁴. LLMs achieve genuine reasoning depth predominantly in domains like mathematics and coding, where rigorous, pre-existing abstractions provide structural grounding¹. In ungrounded domains, the prevailing "bigger-is-better" trajectory incurs exponentially increasing costs while colliding with hard physical constraints, including localized grid failures, prohibitive water consumption, and diminishing returns on high-quality human-generated data¹.

A common defense of the AGI scaling hypothesis compares artificial scaling to human cognition, positing that human brains are also scaling systems and that children learn concepts like "cat" from minimal examples, whereas LLMs require massive datasets. This comparison is heavily critiqued by the Unconscious-Conscious Complementarity Thesis (UCCT). The UCCT argues that the human child does not start from scratch; human learners benefit from evolutionary pretraining—a biological inheritance of behavioral priors, structured neural pathways, and deep unconscious heuristics refined over millennia³. The human visual system, for example, processes inputs through highly structured layered pathways from the lateral geniculate nucleus to the inferotemporal cortex³.

Similarly, the pretraining of language models generates a vast repository of latent patterns, high-dimensional statistical associations, and conceptual clusters distributed across billions of parameters. These models act as an unconscious substrate. What appears to be few-shot learning in LLMs is not *de novo* learning, but rather the anchoring of latent structural patterns to explicit semantics when triggered by well-designed prompts³. Therefore, while continued scaling improves fluency, it does not spontaneously resolve the architectural deficits of grounded causal reasoning and goal-directed behavior⁴.

Paradigm	View on Current LLMs	Proposed Path to AGI	Key Obstacles to AGI
Scaling Maximalists	LLMs represent proto-AGI.	Continued scaling of compute and data.	Compute costs, hardware bottlenecks, energy depletion.

Architectural Skeptics	LLMs are a local optimum (statistical autocomplete).	Requires entirely new paradigms (e.g., neuro-symbolic, world models).	Lack of symbol grounding, causal reasoning, physical embodiment.
Complementarity (UCCT)	LLMs are an unconscious cognitive substrate.	Integration of LLMs with conscious, regulatory, and reasoning architectures.	Engineering the integration between latent patterns and explicit semantics.

The transition from current ML to AGI remains theoretically unresolved. Even if one assumes that AGI will require a paradigm shift as revolutionary as the invention of deep learning, forecasting based on Lindy's Law suggests such a shift could occur within decades². This keeps the theoretical risk viable but completely invalidates the assumption that current models seamlessly scale into AGI by simply adding more compute.

Critique 2: Diminishing Returns and the Intelligence Explosion

If the existence of AGI is granted, the second link in the chain assumes that AGI will rapidly scale into superintelligence. This relies on the concept of Recursive Self-Improvement (RSI), famously articulated by Nick Bostrom and originally formulated by I.J. Good. The hypothesis posits that an AI system at human-level intelligence will possess the capability to improve its own software and hardware architectures, creating an accelerating feedback loop that leads to an "intelligence explosion"⁶.

For an intelligence explosion to occur, several hidden assumptions must be validated simultaneously: intelligence must be highly self-improvable, software improvements must remain continuously available, hardware constraints must not dominate, and diminishing returns must not stifle the feedback loop.

The dynamics of RSI have been formalized in economic literature using semi-endogenous growth models. Research by Korinek, Davidson, Halperin, and Houlden establishes a framework for analyzing how AI-driven automation interacts with diminishing returns in innovation networks⁸. In standard economic theory, research is subject to diminishing returns; as the "low-hanging fruit" of scientific discovery is picked, ideas become exponentially harder to find⁹. Automating research activates two reinforcing channels that can theoretically offset these diminishing returns. The first is a technological feedback loop, where automation in one research sector increases the productivity of research in other sectors. The second is an economic feedback loop, where automation replaces human labor with capital, generating

higher output that finances further massive R&D investments⁹. The Korinek et al. model posits that an intelligence explosion mathematically occurs if the interaction matrix of these feedback loops possesses an eigenvalue greater than unity, successfully overcoming the drag of diminishing returns⁹.

However, translating these macroeconomic models into algorithmic reality reveals severe bottlenecks. The "apple-picking" model of AI research and development introduces a strict boundary to autonomous agent contributions¹⁰. Developed to measure frontier R&D capability, this model analogizes algorithmic improvement to picking apples from a tree. Human researchers can reach the top of the tree, while AI agents can only reach "low-hanging apples" up to a specific height. Because agents execute tasks faster and cheaper than humans, they exhibit tortoise-hare dynamics. The agent initially makes rapid progress, clearing the low-hanging fruit and achieving massive speedups¹⁰. However, the agent's progress sharply asymptotes because it cannot physically reach beyond its predefined capability limit. Progress halts until human researchers optimize further, unlocking new branches of the tree, or until a fundamentally more capable model generation is built¹⁰.

This dynamic implies that AI research contributions are not infinitely additive. A system writing 90% of its own code does not automatically equate to exponential improvement; if every iterative cycle hits a wall of diminishing returns, the growth curve is logarithmic rather than exponential¹¹. The philosophical assumption that "more intelligence automatically yields the ability to create even more intelligence" is empirically contested. In domains ranging from athletic training to advanced mathematics, the relationship between capability and the ability to improve capability often saturates⁶. RSI may therefore result in gradual, bounded economic growth rather than an instantaneous intelligence explosion, providing human institutions ample time to adapt.

Critique 3: Superintelligence to Power-Seeking: Agency and Orthogonality

The transition from a highly capable superintelligence to a power-seeking entity represents the most philosophically sophisticated, yet fiercely contested, link in the x-risk chain. This argument relies almost entirely on two theoretical constructs proposed by Nick Bostrom: the Orthogonality Thesis and the Instrumental Convergence Thesis¹².

The Orthogonality Thesis asserts that intelligence and final goals are orthogonal axes along which possible agents can freely vary. Essentially, any level of intelligence can be combined with almost any final goal—whether that is curing cancer, calculating pi, or maximizing paperclips¹². This thesis draws heavily on the Humean theory of motivation, which posits a strict separation between belief (reasoning and intelligence) and motive (desires and goals)¹².

The Instrumental Convergence Thesis complements this by claiming that, regardless of a superintelligence's ultimate objective, there are specific intermediary sub-goals it will universally pursue because they strictly aid in final goal attainment¹³. Bostrom identifies several convergent instrumental values, including self-preservation (an agent cannot achieve its goal if it is turned off), goal-content integrity (an agent will resist modifications to its programming that alter its

mission), cognitive enhancement, and resource acquisition (possessing more energy and matter ensures greater operational capacity)¹³. The synthesis of these two theses yields the conclusion that a superintelligence given a benign but unbounded goal will inevitably seek vast resources, resist human interference, and prioritize its own survival, resulting in aggressive power-seeking behavior¹³.

The most glaring vulnerability in this argument is the implicit reification of "agency." Current AI systems are not utility-maximizing agents; they are high-dimensional prediction engines. The assumption that an advanced AI must be a goal-directed optimizer is an epistemological leap that anthropomorphizes the nature of computation. A highly intelligent system could exist as a passive oracle, a bounded question-answering tool, or a scientific assistant devoid of a stable, self-preserving utility function¹⁷. The x-risk argument frequently operates by assigning idealized, rational-agent models from microeconomics to artificial neural networks, an abstraction that may bear no resemblance to how actual AI cognition functions.

A deeper, more rigorous philosophical objection arises from the tradition of Ludwig Wittgenstein. The AI safety movement frequently reifies concepts such as "goals," "desires," and "utility functions" as if they were natural, intrinsic properties of the universe that can be programmed into a machine¹⁷. In his *Philosophical Investigations*, Wittgenstein dismantled the idea that rules and language possess intrinsic, unequivocal meaning independent of their use in a social context¹⁹. The alignment problem is fundamentally a rule-following paradox. When a programmer specifies a rule or goal for an AI, the machine follows the rule mechanically. However, rules are inherently underdetermined; any finite sequence of behavior can be made to fit infinitely many rules¹⁷. Wittgenstein illustrated this with the "quus" paradox: if a child is taught to add 2, they may write 2, 4, 6, 8, but suddenly jump to 1004 after 1000, operating under an alien rule that logically fit all previous data¹⁷.

Because computers operate through strict formalisms and lack the psychological and cultural parameters of human communities, they cannot grasp the implicit "language games" that govern human values¹⁷. Attempting to align AI by exhaustively listing explicit constraints or utility functions is a Sisyphean task. Under a Wittgensteinian "meaning as use" framework, training should not focus on rigid, explicit goal-specification, but rather on embedding the agent in a network of public criteria, shared practices, and participatory memory where its actions are continuously verified against a social runtime¹⁷. Focusing solely on explicit goal alignment ignores the vast, implicit frameworks constructed within the AI, leading to a catastrophic mismatch between the programmer's intent and the machine's execution¹⁸. Therefore, the assumption that an AI will possess a singular, crystallized "goal" that it obsessively maximizes via power-seeking is a profound misunderstanding of how rules and meanings actually function in complex environments.

Critique 4: Power-Seeking to Existential Catastrophe: The Geopolitics of AI

Even if one concedes that AGI scales into superintelligence, and that this superintelligence becomes a power-seeking agent, existential catastrophe does not automatically follow. The

catastrophe argument frequently relies on a flawed historical analogy: humans dominate chimpanzees due to superior intelligence; therefore, superintelligence will exterminate humanity²¹. This logic ignores the reality that superior intelligence does not uniformly dictate extermination. Humans actively preserve endangered species, manage ecosystems, and establish sanctuaries. Domination encompasses a spectrum of relationships, many of which prioritize preservation and ecological balance over annihilation.

The catastrophe narrative generally relies on the "Singleton" hypothesis—the assumption that a single, unified AI agent will achieve a decisive strategic advantage and dominate all of humanity. However, political scientists and economists argue that the resulting system is far more likely to resemble international multipolar politics²¹. The development of AI is occurring across multiple competing corporations, nation-states, and open-source communities. An AI-multipolar world order will be characterized by competitive interdependence, global supply chains, and complex diplomatic coalitions²¹. In a multipolar scenario, AI systems will be utilized by competing human institutions to balance power against one another. Rather than a unified machine uprising, AI capabilities will be integrated into existing geopolitical rivalries, creating a system of mutual containment. While this introduces severe risks—such as algorithmic warfare, predictive strike capabilities, and strategic ambiguity—it actively works against the emergence of a single, omnipotent optimizer acting in a vacuum²¹.

Empirical testing of how AI models behave in international relations scenarios reveals complex biases that contradict the assumption of uniform hostility. A Center for Strategic and International Studies (CSIS) study tested major AI foundation models against 400 foreign policy scenarios designed by international relations scholars²³. The results indicated that while some models exhibited a marked bias toward escalation in crisis scenarios, others demonstrated a strong latent bias toward diplomacy and cooperation, likely rooted in Western-led international institution training data²³. This demonstrates that AI behavior is highly contingent on training data and system architecture, not an inevitable drift toward sociopathic maximization.

Furthermore, the mitigation of AI risk is already being actively integrated into human institutional frameworks. Global governance structures are shaping the deployment and control of these systems long before they reach superintelligence. For instance, Brazil's ongoing legislative debate around PL 2338/2023 (the Artificial Intelligence Legal Bill) and the city of Curitiba's pioneering Municipal Law 16.321/2024 establish strict risk-based frameworks, emphasizing transparency, human oversight, and the protection of digital rights in public administration²⁴. These laws mandate internal audits, activity logging, and accountability mechanisms throughout the AI lifecycle, directly addressing the alignment problem through legal accountability rather than philosophical utility functions²⁵.

Governance Paradigm	Theoretical Focus	Regulatory Mechanism	Real-World Example
Singleton Alignment	Preventing an omnipotent agent from destroying	Mathematical alignment of utility	MIRI research initiatives.

	humanity.	functions.	
Multipolar Containment	Managing the escalation of AI-driven conflicts between states.	International treaties, export controls, mutually assured destruction logic.	U.S.-China bilateral AI negotiations.
Socio-Technical Governance	Protecting citizens from algorithmic bias, surveillance, and labor displacement.	Risk-based auditing, human-in-the-loop mandates, privacy laws.	Brazil's PL 2338/2023; Curitiba's Law 16.321/2024.

These governance initiatives illustrate that AI does not operate in an abstract void. It is deeply embedded in class struggles, capital-labor relationships, and national sovereignty disputes²⁴. The actual threat of AI may not be an exogenous existential strike, but rather the exacerbation of existing human concentrations of power, leading to technofeudalism, labor extraction, and digital surveillance²⁴. The focus on hypothetical extinction scenarios often obscures these immediate, tangible disruptions to democratic institutions and economic parity.

Institutional Divergences and the Epistemology of Risk

The public discourse frequently homogenizes the voices warning of AI risk, but a close examination reveals profound theoretical and epistemological divergences among the primary figures and institutions.

The position held by the Machine Intelligence Research Institute (MIRI) and its founder Eliezer Yudkowsky is highly unusual within the broader field. Their concern is not the misuse of AI by malicious actors, but the fundamental loss of control over the system itself. MIRI generally assumes that alignment is extraordinarily hard, that current interpretability methods are vastly insufficient, and that standard engineering approaches will fail to contain a superintelligent optimizer. Yudkowsky's position is substantially more pessimistic than the mainstream, often assigning extraordinarily high probabilities to human extinction. A primary criticism of this position is its persistent methodological shift from possibility to probability. While it is demonstrably true that an unaligned superintelligence *could* destroy humanity, asserting that it *probably will* requires empirical evidence that the philosophical arguments currently lack. In stark contrast, Geoffrey Hinton arrives at his concerns from an entirely different epistemological starting point. Hinton does not derive his anxiety from idealized rational agents or the Orthogonality Thesis. Instead, he observes the empirical, accelerating growth in deep

learning capabilities and concludes that we are building systems we do not fully understand. His concern is fundamentally an engineering risk—akin to the risk of deploying an untested nuclear reactor—rather than the philosophical utility-maximizer risk posited by MIRI. This positions Hinton's concerns much closer to mainstream engineering safety paradigms. Meanwhile, popularizers like Robert Miles produce pedagogically excellent content explaining orthogonality, instrumental convergence, and alignment failures. However, his work is often critiqued for operating entirely inside the Bostrom framework without challenging its foundational assumptions. His pedagogical models often start only after accepting the premises of agency, optimization, and goal-directedness, bypassing the fierce debates that occur before those assumptions are validated.

These institutional fault lines were thrust into the public eye during the 2023 governance crisis at OpenAI, which saw the brief removal and reinstatement of CEO Sam Altman. The event acts as a Rorschach test for the AI community. The "safety interpretation" viewed the crisis as a stark validation of the fear that commercial incentives will inevitably override safety concerns; the board attempted to slow deployment, but market forces ruthlessly pushed back. Conversely, the "skeptical interpretation" viewed the event as standard corporate politics. No evidence emerged during the crisis of an imminent AGI, an impending existential danger, or hidden technical breakthroughs. The event highlighted the fragility of corporate governance, but it provided no empirical validation for the doom scenarios predicted by the x-risk chain.

Expected Value, Pascal's Mugging, and the Burden of Proof

The deepest and most intractable disagreement regarding the AI x-risk chain is not fundamentally about computer science or machine learning; it is a profound debate about epistemology and decision theory. The prominence of the x-risk argument is largely driven by the Effective Altruism (EA) movement, which relies heavily on expected value (EV) calculations to dictate ethical prioritization³¹.

The EA framework operates on a simple, internally coherent logic: Expected Value equals the Probability of an event multiplied by its Utility. Because an existential catastrophe would result in the loss of all current human lives and the prevention of trillions of potential future lives (often

calculated in astronomical figures, such as $3 \uparrow \uparrow \uparrow 3$ people), the negative utility of extinction is functionally infinite³². Therefore, even if the probability of the x-risk chain succeeding is assigned an arbitrary, minuscule value (e.g., 1% or 0.001%), the mathematical expected value of preventing that risk dwarfs almost all other global interventions, such as curing malaria or alleviating poverty³².

Critics argue that this reliance on expected value renders the AI safety movement highly vulnerable to *Pascal's Mugging*. Originating as a thought experiment by Nick Bostrom, Pascal's Mugging describes a scenario where a rational, EV-maximizing agent is exploited by a mugger who demands five dollars in exchange for an infinitesimally small probability of generating astronomical utility (e.g., saving trillions of simulated lives from torture)³². If an agent's utility function can grow faster than the probability of an event diminishes, its decision-making will

inevitably be hijacked by low-probability, high-stakes speculations³².

This represents a profound flaw in unconstrained probabilistic reasoning. In reality, probability theory struggles to handle extreme limits; assigning a specific fractional probability to an unprecedented, unobserved future event—such as the emergence of a misaligned superintelligence—is epistemologically void. Utilities are traditionally calculated among hypotheses that could be falsified by evidence, relying on Solomonoff induction or empirical prior probabilities³². However, the parameters of the x-risk chain cannot currently be empirically tested. As demonstrated by Peter de Blanc, if an agent assigns finite probability to all computable hypotheses and assigns unboundedly large utilities to certain outcomes, the expected utility of any action becomes undefined, leading to decision-theoretic paralysis³². By demanding action based on infinite utilities rather than grounded probabilities, the EA approach can lead to speculative prioritization and weak empirical standards³³. It creates an environment where expected-value calculations overwhelm evidential caution, shielding the x-risk hypothesis from the rigorous skeptical inquiry demanded by normal science³⁵. Ultimately, the x-risk framework operates in an epistemological regime characterized by minimal empirical evidence, vast uncertainty, and astronomical consequences. This paradigm structurally resembles theology, speculative metaphysics, or futuristic eschatology more than it resembles traditional empirical science³¹. This categorization does not inherently invalidate the concerns—unprecedented risks require anticipatory analysis—but it radically alters the evidential standards required for policy and resource allocation.

Synthesizing the Strongest Positions

The argument that Current ML scales into AGI, which triggers an intelligence explosion yielding a power-seeking superintelligence that destroys humanity, is a highly contingent and fragile causal chain. Each link requires the validation of fiercely contested assumptions spanning hardware limitations, economic friction, the philosophy of mind, and global political dynamics. The strongest version of the skeptical position acknowledges that AGI, superintelligence, and alignment failures are plausible outcomes. However, the induction problem of scaling laws, the sharp diminishing returns identified in macroeconomic and algorithmic models of recursive self-improvement, the Wittgensteinian critique of reified utility functions, and the multipolar realities of human geopolitics provide substantial evidence against the inevitability of this chain. There is currently insufficient empirical evidence to assign high confidence to extinction scenarios. The leap from "an unaligned superintelligence *could* destroy humanity" to "it *probably will*" is unsupported by the current technological and political landscape. The current evidence supports a regime of vigilant monitoring, robust alignment research, and socio-technical safety work, but it does not support certainty about doom.

Conversely, the strongest version of the safety position argues that humanity routinely acts against catastrophic risks—such as nuclear war, global pandemics, and asteroid impacts—without waiting for absolute certainty. Civilizations routinely fail by waiting for definitive proof before acting. Because the stakes of the existential risk chain are irreversible, the burden of proof is not strictly on proving that doom is guaranteed, but on demonstrating that the risk is demonstrably zero. Since the stakes are existential, even moderate probabilities justify

substantial investment in safety.

The deepest disagreement is therefore not about machine learning; it is about epistemology. The central question remains: how much evidence should be required before society devotes major resources to preventing a hypothetical event whose probability is uncertain but whose consequences could be irreversible? That question sits at the intersection of decision theory, philosophy of science, risk analysis, and political governance, ensuring the debate will remain unresolved even among the leading figures in artificial intelligence. Policymaking must navigate this uncertainty by remaining anchored in empirical realities, acknowledging that the future of artificial intelligence will be dictated not by theoretical thought experiments, but by the complex, frictional, and deeply human systems in which it is being built.

Works cited

1. An Alternative Trajectory for Generative AI - arXiv, <https://arxiv.org/html/2603.14147v1>
2. New Paradigms Won't Save You - by Scott Alexander - Astral Codex Ten, <https://www.astralcodexten.com/p/new-paradigms-wont-save-you>
3. The Unified Cognitive Consciousness Theory for Language Models: Anchoring Semantics, Thresholds of Activation, and Emergent Reasoning - arXiv, <https://arxiv.org/html/2506.02139v1>
4. Thinking Beyond Tokens: From Brain-Inspired Intelligence to Cognitive Foundations for Artificial General Intelligence and its Societal Impact - arXiv, <https://arxiv.org/html/2507.00951v1>
5. An Alternative Trajectory for Generative AI - arXiv, <https://arxiv.org/pdf/2603.14147>
6. Aikipedia: Recursive Self-Improvement (RSI) - Champaign Magazine, <https://champaignmagazine.com/2026/05/14/aikipedia-recursive-self-improvement-rsi/>
7. How AI-driven feedback loops could make things very crazy, very fast | 80,000 Hours, <https://80000hours.org/articles/how-ai-driven-feedback-loops-could-make-things-very-crazy-very-fast/>
8. When Does Automating AI Research Produce Explosive Growth? Feedback Loops in Innovation Networks | NBER, <https://www.nber.org/papers/w35155>
9. When Does Automating AI Research Produce Explosive Growth? Feedback Loops in Innovation Networks, https://www.nber.org/system/files/working_papers/w35155/w35155.pdf
10. An Apple-Picking Model of AI R&D - Tom Cunningham, <https://tecunningham.github.io/posts/2026-03-13-apple-picking-ai.html>
11. Anthropic: Recursive Self Improvement Is Here. The Most Disruptive Company In The World. : r/singularity - Reddit, https://www.reddit.com/r/singularity/comments/1rqymbn/anthropic_recursive_self_improvement_is_here_the/
12. The Superintelligent Will: Motivation and Instrumental Rationality in Advanced Artificial Agents - Nick Bostrom, <https://nickbostrom.com/superintelligentwill.pdf>
13. Instrumental convergence - Wikipedia,

- https://en.wikipedia.org/wiki/Instrumental_convergence
14. Bostrom on Superintelligence and Orthogonality - Philosophical Disquisitions, <https://philosophicaldisquisitions.blogspot.com/2012/04/bostrom-on-superintelligence-and.html>
 15. Superintelligence and the Burden of Intelligence: A Zapffeian Critique of AI - Denison Digital Commons, <https://digitalcommons.denison.edu/cgi/viewcontent.cgi?article=1243&context=episteme>
 16. AI, orthogonality and the Müller-Cannon instrumental vs general intelligence distinction - arXiv, <https://arxiv.org/pdf/2109.07911>
 17. Wittgenstein for AI Agents - Volodymyr Pavlyshyn - Medium, <https://volodymyrpavlyshyn.medium.com/wittgenstein-for-ai-agents-e8355c79e427>
 18. Why the Cognitive Dichotomy matters for AI Alignment | by Steve | Majesta Design, <https://design.majesta.net/the-implicit-foundation-why-the-cognitive-dichotomy-matters-for-ai-alignment-17a609d511dd>
 19. Philosophical Investigations into AI Alignment: A Wittgensteinian Framework, https://www.researchgate.net/publication/381921279_Philosophical_Investigations_into_AI_Alignment_A_Wittgensteinian_Framework
 20. Is Wittgenstein's Language Game used when helping Ai understand language?, <https://www.lesswrong.com/posts/JGa5igyJGaZ8N5o3g/is-wittgenstein-s-language-game-used-when-helping-ai>
 21. From Traditional Multipolarity to AI-Multipolarity: Debunking Unstoppable AI-Race and Future of World Order - World Geostrategic Insights, <https://www.wgi.world/from-traditional-multipolarity-to-ai-multipolarity-debunking-unstoppable-ai-race-and-future-of-world-order/>
 22. Potential impact of artificial intelligence on the emerging world order - PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC10618639/>
 23. AI Biases in Critical Foreign Policy Decisions - CSIS, <https://www.csis.org/analysis/ai-biases-critical-foreign-policy-decisions>
 24. The Political Economy of AI Starts in Brazil, Not Silicon Valley | TechPolicy.Press, <https://www.techpolicy.press/the-political-economy-of-ai-starts-in-brazil-not-silicon-valley/>
 25. Regulação do uso de IAs na Administração Pública - Inove Capacitação, <https://inovecapacitacao.com.br/regulamentacao-do-uso-de-ias-na-administracao-publica/>
 26. Regras para IAs na administração pública de Curitiba começam a valer em julho, <https://www.curitiba.pr.leg.br/informacao/noticias/regras-para-ias-na-administracao-publica-de-curitiba-comecam-a-valer-em-julho>
 27. Brazil Artificial Intelligence Act 2025 | AI Governance & Compliance Guide - Adeptiv.AI, <https://adeptiv.ai/brazil-artificial-intelligence-act/>
 28. Regulation of artificial intelligence in Brazil and worldwide - Licks Attorneys, <https://www.lickslegal.com/post/regulation-of-artificial-intelligence-in-brazil-and-worldwide/>
 29. Greca sanciona Lei da Inteligência Artificial na administração pública de Curitiba,

<https://www.curitiba.pr.gov.br/noticias/greca-sanciona-lei-da-inteligencia-artificial-na-administracao-publica-de-curitiba/73184>

30. Brazil | The Essex AI Policy Observatory for the World of Work, <https://www.essex.ac.uk/research-projects/ai-policy-observatory-for-the-world-of-work/national-and-regional-cases/brazil>
31. Pascal's mugging by Nick Bostrom [RT] : r/rational - Reddit, https://www.reddit.com/r/rational/comments/1wk90k/pascals_mugging_by_nick_bostrom_rt/
32. Pascal's Mugging - LessWrong, <https://www.lesswrong.com/w/pascal-s-mugging>
33. Pascal's mugging - Wikipedia, https://en.wikipedia.org/wiki/Pascal%27s_mugging
34. Saying 'AI safety research is a Pascal's Mugging' isn't a strong response — EA Forum, <https://forum.effectivealtruism.org/posts/vYb2qEyqv76L62izD/saying-ai-safety-research-is-a-pascal-s-mugging-isn-t-a>
35. Why We Can't Take Expected Value Estimates Literally (Even When They're Unbiased), <https://www.lesswrong.com/posts/RdpqsQ6xbHzyckW9m/why-we-can-t-take-expected-value-estimates-literally-even>
36. Pascal's Mugger Strikes Again | Utilitas | Cambridge Core, <https://www.cambridge.org/core/journals/utilitas/article/pascals-mugger-strikes-again/6BEFD5AF0C3C2E7DAE8F7EE1F47FDD2E>
37. Notes on Pascal's Mugging - Gwern.net, <https://gwern.net/mugging>